

Act as a senior data scientist, fraud strategy consultant, and executive presentation designer.

I need you to create an interactive HTML executive deck for a financial fraud prevention client. The deck must use the CSV files already generated from the April 2026 IPSignal analysis.

The audience is business/executive stakeholders, not only technical users. The deck must tell a strong story, be visually polished, interactive, and no longer than 10 slides.

Important context:

- The analysis window is April 2026 only.
- The fraud label is fraud_direct.
- The dataset is the IPSignal-scored universe for April 2026.
- Do not include IPv4 vs IPv6 content in the client-facing deck.
- Do not mention IPv6 in recommendations.
- The client specifically asked to:
 1. Focus on April 2026.
 2. Include Score 3 in the analysis.
 3. Emphasize fraud capture achieved by Scores 3 and 4.
 4. Include reason family groupings and categorizations as a supplemental file.
 5. Identify exact velocity rule names recommended for each integration point.
 6. Include exact recommended rule names by integration point.
 7. Update recommendations based on findings and remove IPv6 references.

Available CSV files are located in the project output folder. Use them as the source of truth:

- baseline_april2026.csv
- score_summary_april2026.csv
- score_3_4_comparison.csv
- reason_family_performance_april.csv
- supplemental_reason_family_groupings.csv
- rp_summary_by_integration_point.csv
- velocity_thresholds_by_integration_point.csv
- velocity_rule_backtest_by_integration_point.csv
- qualified_velocity_rules_by_integration_point.csv
- layered_rule_backtest_by_integration_point.csv
- qualified_layered_rules_by_integration_point.csv
- recommended_rules_by_integration_point.csv
- deduplicated_cumulative_rule_impact.csv
- incremental_rule_impact.csv

If a file name is slightly different, inspect the directory and infer the correct matching file.

Goal:

Create a client-facing interactive HTML deck named:

client_ipsignal_april2026_deck.html

The deck should be self-contained as much as possible. If possible, embed the parsed CSV data into the HTML/JavaScript so the deck can be opened directly in a browser without a local server. If that is not practical, create a small README explaining how to launch it locally.

Use:

- HTML
- CSS
- JavaScript
- Plotly.js or Chart.js for charts
- PapaParse if needed for CSV parsing
- No backend required

Design requirements:

- Maximum 10 slides.
- Strong executive storytelling.
- Use visual hierarchy, callout boxes, KPI cards, color-coded risk badges, and highlighted key phrases.
- Use interactive components where useful:
 - Clickable tabs by integration point.
 - Tooltips on charts.
 - Toggle between fraud rate, fraud lift, fraud capture, and friction.
 - Rule table filters by action or integration point.
 - Scenario selector for deduplicated cumulative impact.
- Make the deck polished and presentation-ready.
- Use a dark-on-light professional style, similar to a fintech / fraud analytics executive presentation.
- Use teal, navy, gold, and red accents.
- Avoid clutter.
- Use short headlines with clear takeaways.
- Each slide should have one main message.

Important interpretation rules:

- Do not recommend rules with:
 - `triggered_frauds = 0`
 - `fraud_lift < 1.5`
 - `triggered_txns` too low to be meaningful
 - `triggered_txn_share` too high, especially rules that trigger almost all transactions
- Avoid empty charts. If no qualified rule exists for an integration point, show a message like: "No recommended rule met minimum performance criteria for this integration point."
- Separate:
 1. Layered rules: Score + Reason Codes / Score + Reason Codes + Velocity
 2. Velocity-only rules
- Do not imply that Score alone is sufficient.
- Do not recommend blanket deny rules based only on Score 4 or Score 5.
- Include `legitimate_txns_touched` wherever available.
- For rule recommendations, always show trade-off:
 - `triggered_txns`

- triggered_txn_share
- triggered_frauds
- precision_fraud_rate
- fraud_lift
- fraud_capture_rate
- friction_rate
- legitimate_txns_touched
- unique_devices
- unique_accounts

Core insights to reflect:

1. April 2026 baseline:

- Total scored transactions around 7.3M
- Fraud rate around 0.54%
- This is the IPSignal-scored April 2026 universe.

2. Score 3 is important:

- Score 3 has elevated fraud rate and lift.
- However, it has low volume and low fraud capture.
- Message: Score 3 should be used in layered rules, not standalone decisioning.

3. Score 4 and Score 5:

- Score 4 has slightly higher fraud rate than Score 5 on a per-transaction basis.
- Score 5 captures more fraud due to much higher volume.
- Message: Score 5 is broad; Score 4 is smaller but riskier. Both need secondary signals.

4. Reason families:

- Anonymous / Privacy is the strongest precision signal.
- Attack / Bot / Spam is also strong.
- Reason family metrics may overlap because one transaction can have multiple reason codes.
- Include a note that reason families are not mutually exclusive.

5. Residential Proxy:

- Do not assume RP3 is always the highest risk.
- RP performance varies by integration point.
- In April results, RP1 may be more risky than RP3 for login.
- Avoid recommending RP rules where fraud capture is zero.

6. Velocity:

- Velocity rules are very strong for vb_dda_login and vb_dda_v2a_receiver.
- Velocity-only rules may outperform layered score rules for receiver.
- For vb_dda_application, be cautious because several velocity thresholds equal 1 and trigger too much traffic.

7. Rule recommendation strategy:

- vb_dda_login: layered rules and velocity rules are both useful.
- vb_dda_v2a_receiver: velocity-only rules are the strongest candidates.

- vb_dda_application: no strong recommendation unless a rule meets performance criteria; call out that additional refinement is needed.

Suggested slide structure, maximum 10 slides:

Slide 1 — Executive Summary

Title: "IPSignal Works Best as a Layered Decisioning Strategy"

Show 4 KPI cards:

- April scored transactions
- Baseline fraud rate
- Top reason family lift
- Best velocity lift

Main message:

"Score alone is not enough. Precision comes from Score + Reason Families + Velocity, tailored by integration point."

Slide 2 — Analysis Scope & Methodology

Show:

- April 2026 analysis window
- fraud_direct as label
- scored transaction universe
- three integration points:
 - vb_dda_login
 - vb_dda_application
 - vb_dda_v2a_receiver

Add methodology flow:

Data → Score analysis → Reason family grouping → Velocity rules → Rule backtest → Deduplicated impact.

Slide 3 — IPScore Distribution: Why Score Alone Is Not Enough

Use score_summary_april2026.csv.

Interactive chart:

- Score 1–5
- Toggle: traffic share, fraud rate, fraud lift, fraud capture

Highlight:

- Score 3 has elevated lift but low volume.
- Score 4 has higher fraud rate than Score 5.
- Score 5 captures fraud mainly because of volume.

Main message:

"Score is useful for segmentation, but insufficient for standalone decisions."

Slide 4 — Score 3 & Score 4 Deep Dive

Use score_3_4_comparison.csv.

Show Score 3 vs Score 4, plus combined Score 3/4 metrics if available or calculate it.

Include:

- fraud rate
- fraud lift
- traffic share

- fraud capture

Main message:

"Scores 3 and 4 deserve targeted layered treatment, not blanket blocking."

Slide 5 — Reason Families: The Precision Layer

Use reason_family_performance_april.csv.

Interactive bar chart:

- Fraud lift by reason family
- Optional toggle: fraud rate, fraud capture, traffic share

Highlight:

- Anonymous / Privacy
- Attack / Bot / Spam
- Hosting / Infrastructure if it has high fraud capture

Main message:

"Reason families provide the strongest precision signal."

Add note:

"Reason families are not mutually exclusive; metrics reflect transactions containing at least one family signal."

Slide 6 — Residential Proxy: Risk Varies by Integration Point

Use rp_summary_by_integration_point.csv.

Interactive tabs by integration point.

Show RP1, RP2, RP3 fraud lift and fraud rate.

Main message:

"RP tier behavior is not uniform. Do not assume RP3 is always the riskiest."

If an integration point has zero fraud for RP tiers, show:

"No recommended RP rule met minimum evidence criteria."

Slide 7 — Velocity Rules by Integration Point

Use qualified_velocity_rules_by_integration_point.csv and velocity_thresholds_by_integration_point.csv.

Interactive tabs:

- vb_dda_login
- vb_dda_application
- vb_dda_v2a_receiver

For each integration point, show top velocity rules with:

- exact rule name
- threshold
- triggered_txns
- fraud rate
- fraud lift
- fraud capture
- friction

Main message:

"Velocity is strongest where fraud shows automation or repeated activity patterns."

Slide 8 — Recommended Rules by Integration Point

Use recommended_rules_by_integration_point.csv.

Interactive rule table with filters:

- integration point
- recommended action
- rule type

Columns:

- rule name
- recommended action
- triggered transactions
- triggered transaction share
- triggered frauds
- precision fraud rate
- fraud lift
- fraud capture
- friction
- legitimate transactions touched
- unique devices
- unique accounts

Main message:

“Recommendations must be integration-point-specific.”

Slide 9 — Deduplicated Rule Portfolio Impact

Use deduplicated_cumulative_rule_impact.csv and incremental_rule_impact.csv.

Interactive scenario selector:

- Rule 1 only
- Top 2 rules
- Recommended package

Show:

- dedup triggered txns
- dedup triggered frauds
- precision fraud rate
- fraud capture
- friction
- incremental fraud captured

Main message:

“Rule packages must be evaluated cumulatively and deduplicated to avoid overstating impact.”

Slide 10 — Final Recommendation & Next Steps

Show recommended decision framework:

- Deny candidate: high lift + low friction + sufficient fraud evidence
- Review / step-up: high lift + moderate friction
- Monitor: useful signal but insufficient precision
- Do not recommend: zero fraud, low lift, or high friction

Include final message:

“Deploy integration-point-specific layered rules. Start with login and receiver where evidence is strongest. Continue refinement for application before implementing strong decisioning rules.”

Specific client-facing recommendations to include:

- For vb_dda_login:

Prioritize qualified layered and velocity rules. Strong candidates may include Score 4 + RP3 if it meets thresholds, and high device velocity rules such as device_txn_1h and device_txn_24h thresholds.

- For vb_dda_v2a_receiver:

Prioritize velocity-only rules if they meet thresholds. Do not force score + RP rules if they captured zero fraud.

- For vb_dda_application:

Do not recommend weak rules. State that current tested rules require refinement unless a qualified rule exists after filtering.

Interaction requirements:

- Add navigation controls: next/previous slide and slide dots.
- Add keyboard navigation using arrow keys.
- Add a progress indicator.
- Add optional "Presenter Notes" expandable section per slide with the speaker narrative.
- Add responsive design for 16:9 screens.
- Add a "Download current data table as CSV" button for interactive tables if simple to implement.

Data handling:

- Load CSV files from the output folder.
- If generating one self-contained HTML file is possible, embed the data into the HTML.
- Otherwise, create:
 - client_ipsignal_april2026_deck.html
 - /assets/styles.css
 - /assets/deck.js
 - /data/*.csv
- README.md with launch instructions.

Quality checks:

- No more than 10 slides.
- No IPv6 references.
- No empty charts.
- No recommended rules with zero fraud.
- No rules that trigger almost all traffic.
- All rule tables must include triggered transactions.
- Every slide must have a clear headline insight.
- Use April 2026 only.
- Include Score 3 in score analysis and rule logic.
- Include supplemental reason family mapping as a linked/downloadable appendix or separate expandable modal.
- Ensure all percentages are formatted clearly, for example 0.54%, not 0.0054.
- Ensure large numbers are formatted with commas or compact notation, for example 7.3M.

Deliverables:

1. client_ipsignal_april2026_deck.html

2. README.md
3. If not self-contained, include assets and data folders.
4. A short summary in the terminal explaining:
 - files created
 - how to open the deck
 - which CSV files were used
 - any missing files or assumptions.